

Multi-Agent Research Team with Compressed Context Sharing

Project Documentation

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Overview

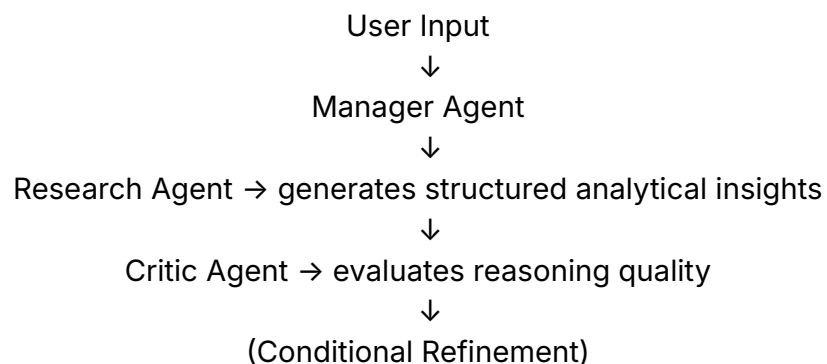
The Multi-Agent Research Support System is a modular AI-driven application designed to demonstrate coordinated multi-agent reasoning using structured orchestration.

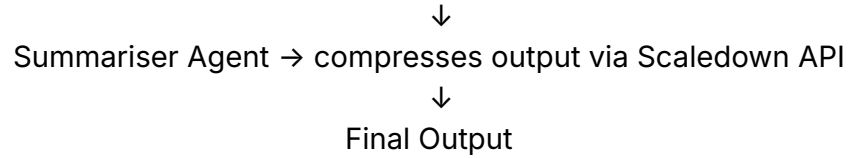
Rather than relying on a single LLM call, the system separates responsibilities across specialised agents:

- Research Agent
- Critic Agent
- Summariser Agent (API-based compression)
- Manager Agent (orchestration layer)

The system showcases how multiple agents collaborate, evaluate outputs, and refine responses through structured coordination.

Architecture:





Agent Responsibilities

1. Research Agent

- Detects topic genre (technical, social, ethical, personal, general)
- Generates structured analytical insights
- Applies domain anchoring for AI-related topics
- Produces assumptions explicitly

Output structure:

```
{  
  "genre": str,  
  "insights": list[str],  
  "assumptions": list[str]  
}
```

2. Critic Agent

- Evaluates analytical depth
- Scores output out of 3
- Checks reasoning coverage
- Verifies domain terminology for technical topics

Output structure:

```
{  
  "score": int,  
  "feedback": list[str]  
}
```

3. Summariser Agent

- Uses Scaledown API for structured compression
- Gracefully falls back if API key is unavailable
- Produces formatted executive summary

API Used:

<https://api.scaledown.xyz/compress/raw/>

Environment Variable Required: **SCALEDOWN_API_KEY**

4. Manager Agent

- Coordinates agent workflow
- Maintains system state
- Triggers refinement loop if critic score is low
- Controls output views

Features

- Modular multi-agent design
- Deterministic genre detection with boundary-safe parsing
- Domain-aware insight generation
- Evaluation-based refinement loop
- API-integrated summarisation
- CLI-based interactive interface
- Error-safe fallback mechanisms
- Clean separation of concerns

Output Includes:

- Analytical insights
- Critic evaluation
- API-compressed summary
- Complete system state (if selected)

Engineering Considerations

- API keys are not hardcoded
- Fallback logic prevents runtime failure
- Input validation ensures non-empty topics
- Genre detection uses boundary-aware regex
- Each agent is independently testable
- Architecture supports future extension (memory agent, retrieval agent, etc.)

Future Improvements

- Retrieval-augmented generation (RAG)
- Persistent conversational memory
- JSON-structured LLM communication
- Web-based interface
- Parallel agent execution

Purpose

This project demonstrates applied multi-agent coordination in generative AI systems. It emphasises structured reasoning, modular design, and evaluation-driven refinement rather than raw LLM output generation.

OUTPUT SCREENS (CLI Interface)

```
PS C:\Users\Lahari\OneDrive\Documents\Intel UNNATI Program\multi-agent-research-team> python main.py

Multi-Agent Research Support System

This system demonstrates coordinated multi-agent reasoning
with research generation, evaluation, and API-based compression.

Enter a topic or idea to explore:
> AI impact on labour markets

Optional: Add scope or focus (press Enter to skip):
> Current global economic trends

Processing exploration...

Choose an output view:
1. Structured research exploration
2. Compressed context summary (API)
3. Critic evaluation
4. Complete system state
5. Exit
> 4
[Manager] Running research agent...
[Manager] Running critic agent...
[Manager] Critic score: 3/3
[Manager] Running summariser agent...
```

```
=====
=== FULL SYSTEM STATE ===

Research Insights:
- In analysing AI impact on labour markets, the dimension of relevance and motivation becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends.
- In analysing AI impact on labour markets, the dimension of primary analytical dimensions becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends.
- In analysing AI impact on labour markets, the dimension of stakeholder perspectives becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends.
- In analysing AI impact on labour markets, the dimension of constraints and risks becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends.
- Recent developments in foundation models, automation of knowledge work, and regulatory debates significantly influence the future trajectory of AI systems.

Critic Score: 3
Critic Feedback:
- Research quality acceptable.

Final Summary:

=== SUMMARY REPORT ===

In analysing AI impact on labour markets, the dimension of relevance and motivation becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends. In analysing AI impact on labour markets, the dimension of primary analytical dimensions becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends. In analysing AI impact on labour markets, the dimension of stakeholder perspectives becomes central. This enables structured reasoning rather than surface-level commentary. The scope is further shaped by the context of Current global economic trends.

[compressed]
```

4th option is the combined result of the results produced by options 1, 2, 3